BIOS vs. UEFI Comparison

	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) tells your computer's OS how to properly operate. It contains instructions on how to control various hardware components.	A modern and more flexible firmware interface that is designed to replace BIOS, acting as a lightweight mini-operating system between platform firmware and OS.
Boot Process	1. Power On – starts CPU executing BIOS 2. BIOS initializes hardware and runs POST 3. BIOS checks boot order and finds a bootable device 4. Loads MBR (Master Boot Record) from the device into memory 5. Transfers control to the MBR bootloader 6. Bootloader loads the OS	1. Power On – CPU starts UEFI firmware 2. Hardware initialization + drivers loaded 3. Boot Device Selection – reads BootOrder from NVRAM 4. Loads EFI bootloader from the EFI System Partition 5. Transfers control to the EFI application / OS bootloader 6. Bootloader loads the OS
Maximum Disk Support	2.2 Terabytes (TB) maximum addressable drive size using standard 512-byte sector sizing.	9.4 Zettabytes (ZB) (approx. 9.4 billion TB) maximum drive size.
Partition Style	MBR (Master Boot Record)	GPT (GUID Partition Table)
Security Features	Minimal firmware-level security; restricted primarily to basic supervisor/user power-on passwords.	Advanced hardware security including Secure Boot (cryptographic key validation)
Speed	Runs 16-bit	Runs 32-bit or 64-bit

Advantages	Widely compatible with older hardware and software	Faster boot, large disk support, better security, graphical UI + mouse, modular drivers, networking support in firmware, multi-boot friendliness.
Disadvantages	Smaller disk support, slow boot, more vulnerable to security threats.	Requires compatibility with the device, there are still flaws with the security.
Modern Usage	Mainly used only for old hardware	Standard on all new PCs

For many years, BIOS has been used as the fundamental firmware that provided computers' operating system with instructions to operate properly. However, as computers started to evolve more with modern technology, such as faster processors, larger storage, and growing security threats, the technology of the BIOS started to get left behind.

BIOS was created by Gary Kildall in 1975, but in 1981, IBM introduced BIOS as a proprietary technology. Years later, Phoenix teamed up with Compaq and Intel to develop the BIOS Boot Specification. As time goes by, its technology started to slow down. One of the biggest limitations was the storage. It was only able to handle 2 terabytes of storage, which in today's world, that is already too small for a lot of job industries. Also, manufacturers started making larger drives and SSDs that BIOS won't be able to manage properly.

As a solution to these problems, UEFI was introduced. It is the much modern version of the BIOS. It works extremely well with larger storage drives without having any problems. UEFI also starts computers much faster than what BIOS does in the past. Most importantly, UEFI includes a security feature called Secure Boot, this checks that the software loading during the startup is trustworthy. This helps with protecting the computer against certain types of attacks that the old BIOS could not.